Yi Xiao

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Profile

Ph.D. student in Computer Science and Technology. Scientific machine learning, data assimilation, and generative modeling for dynamical systems.

Education

Tsinghua University, Beijing, China Ph.D. in Computer Science and Technology, 2022–present (expected summer 2027). Advisor: Prof. Wei Xue. B.S. in Mathematics and Physics, 2018–2022.

Research interests

Scientific machine learning; AI for Science; generative models for dynamical systems; scientific inverse problems; data assimilation.

Research experience

Mohamed bin Zayed University of Artificial Intelligence, October 2025–May 2026 WindINR: latent-state implicit neural representations for local wind query and correction. Project mentors and collaborators: Xiaosong Ma, Pascal Fua, and Robert Jenssen.

Mohamed bin Zayed University of Artificial Intelligence, October 2025–February 2026 MonodVI: mechanistically constrained deep generative modeling of spliced and unspliced RNA. Project mentors and collaborators: Xiaosong Ma and Le Song. Incorporated a steady-state Monod likelihood into variational inference to learn cell-level representations from paired RNA measurements.

Tsinghua University and Shanghai AI Lab, November 2023–October 2025 Joint research on FengWu-4DVar, VAE-Var, and LoRA-EnVar. Project mentors and collaborators: Wei Xue, Lei Bai, and Ben Fei. Developed differentiable forecast–assimilation systems, non-Gaussian generative priors, and parameter-efficient ensemble adaptation.

Tsinghua University, HPCA Lab, January 2021–November 2023 DNN-integrated ensemble data assimilation, advised by Wei Xue. Developed neural surrogates for ensemble perturbation generation to accelerate assimilation workflows.

Publications

See the publications page for selected papers, preprints, and conference abstracts.

Academic service

Reviewer for ICML 2025, NeurIPS 2025, ICLR 2026, and ICML 2026 (**Gold Reviewer**).

Awards and honors

EGU Outstanding Student and PhD Candidate Presentation Award, 2025.

China National Scholarship, 2024–2025 (awarded December 2024).

Hesheng Award for Comprehensive Excellence, 2020–2021.

Hengda Award for Comprehensive Excellence, 2019–2020.

Academic Excellence Award, 2018–2019.

Technical skills and languages

Programming: Python, C/C++, Java, MATLAB. Parallel computing: MPI, OpenMP, CUDA. Languages: Mandarin (native), English (proficient).